Data Compression

- What's wrong with symbol codes?
- Arithmetic Coding

Practical data compression

Claims:

1. The *Shannon information content* of an outcome

$$h(x=a_i) = \log_2 \frac{1}{P(x=a_i)}$$

is a sensible measure of information content.

2. The entropy

$$H(X) = \sum_x P(x) \log_2 \frac{1}{P(x)}$$

is a sensible measure of expected information content.

3. Source coding theorem –
N outcomes from a source X can be compressed into roughly $NH(X)$ bits.

Symbol codes

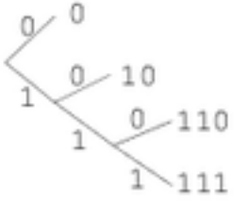
i	a_i	p_i
1	a	0.0575
2	b	0.0128
3	c	0.0263
4	d	0.0285
5	e	0.0913
6	f	0.0173
7	g	0.0133
8	h	0.0313
9	i	0.0599
10	j	0.0006
11	k	0.0084
12	l	0.0335
13	m	0.0235
14	n	0.0596
15	o	0.0689
16	p	0.0192
17	q	0.0008
18	r	0.0508
19	s	0.0567
20	t	0.0706
21	u	0.0334
22	v	0.0069
23	w	0.0119
24	x	0.0073
25	y	0.0164
26	z	0.0007
27	–	0.1928

```
a 0000
b 001000
c 00101
d 10000
e 1100
f 111000
g 001001
h 10001
i 1001
j 1101000000
k 1010000
l 11101
m 110101
n 0001
o 1011
p 111001
q 110100001
r 11011
s 0011
t 1111
u 10101
v 11010001
w 1101001
x 1010001
y 101001
z 1101000001
– 01
```

Last time

- Symbol codes
  - unique decodeability — requires $\sum_i 2^{-l_i} \leq 1$ Kraft inequality

    [if equality holds, the code is 'complete']

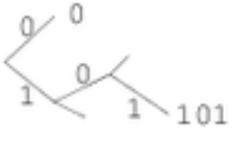


  - prefix codes
    - and binary trees

  - whenever a code achieved L=H, it had

    codelengths l_i^* equal to the information contents

    $$l_i^* = \log \frac{1}{p_i}$$



  - optimal symbol codes

    The optimal symbol code's expected length L satisfies

    $$H(X) \leq L < H(X) + 1$$

  - Huffman algorithm

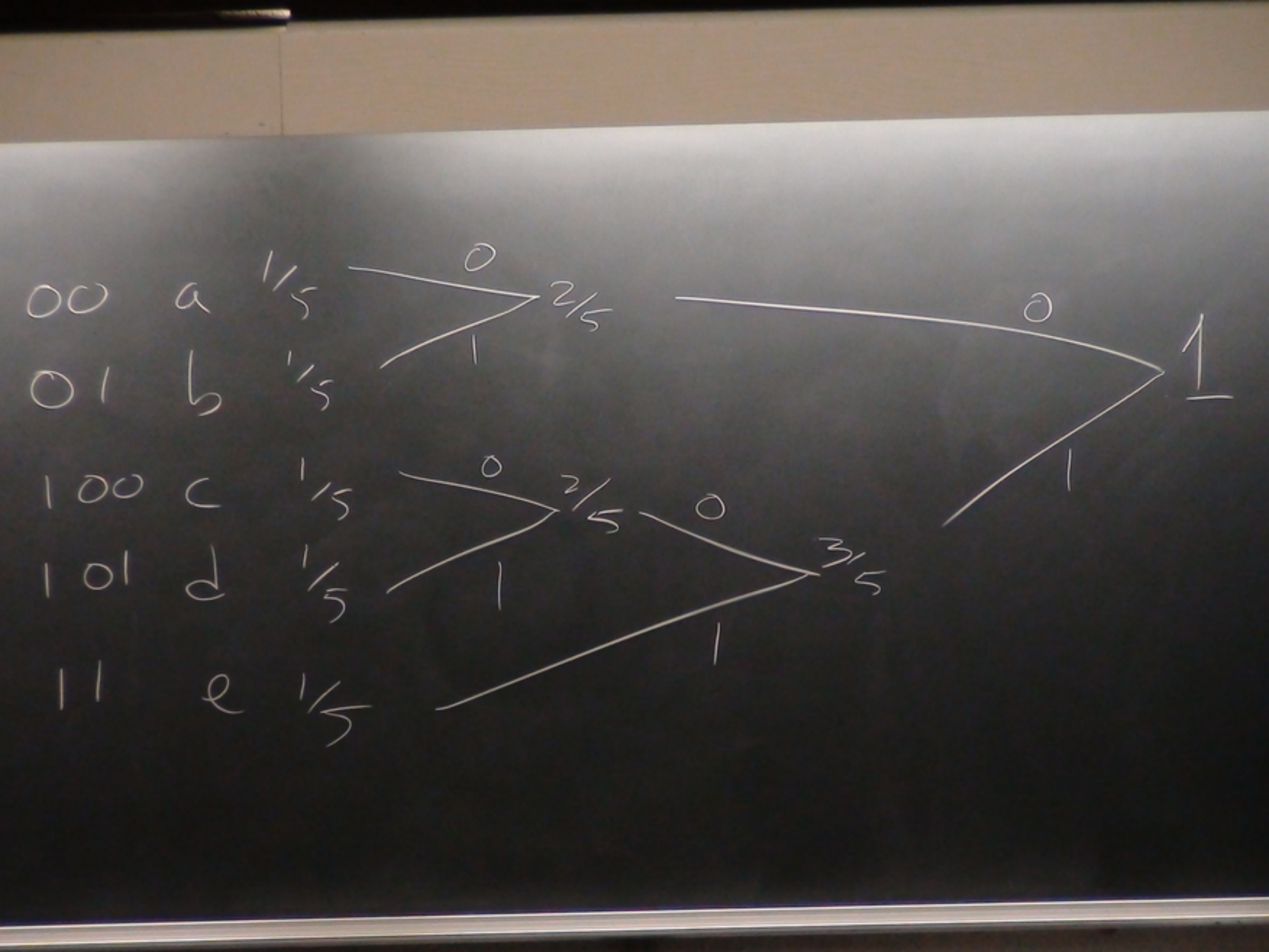
What is the optimal symbol code for

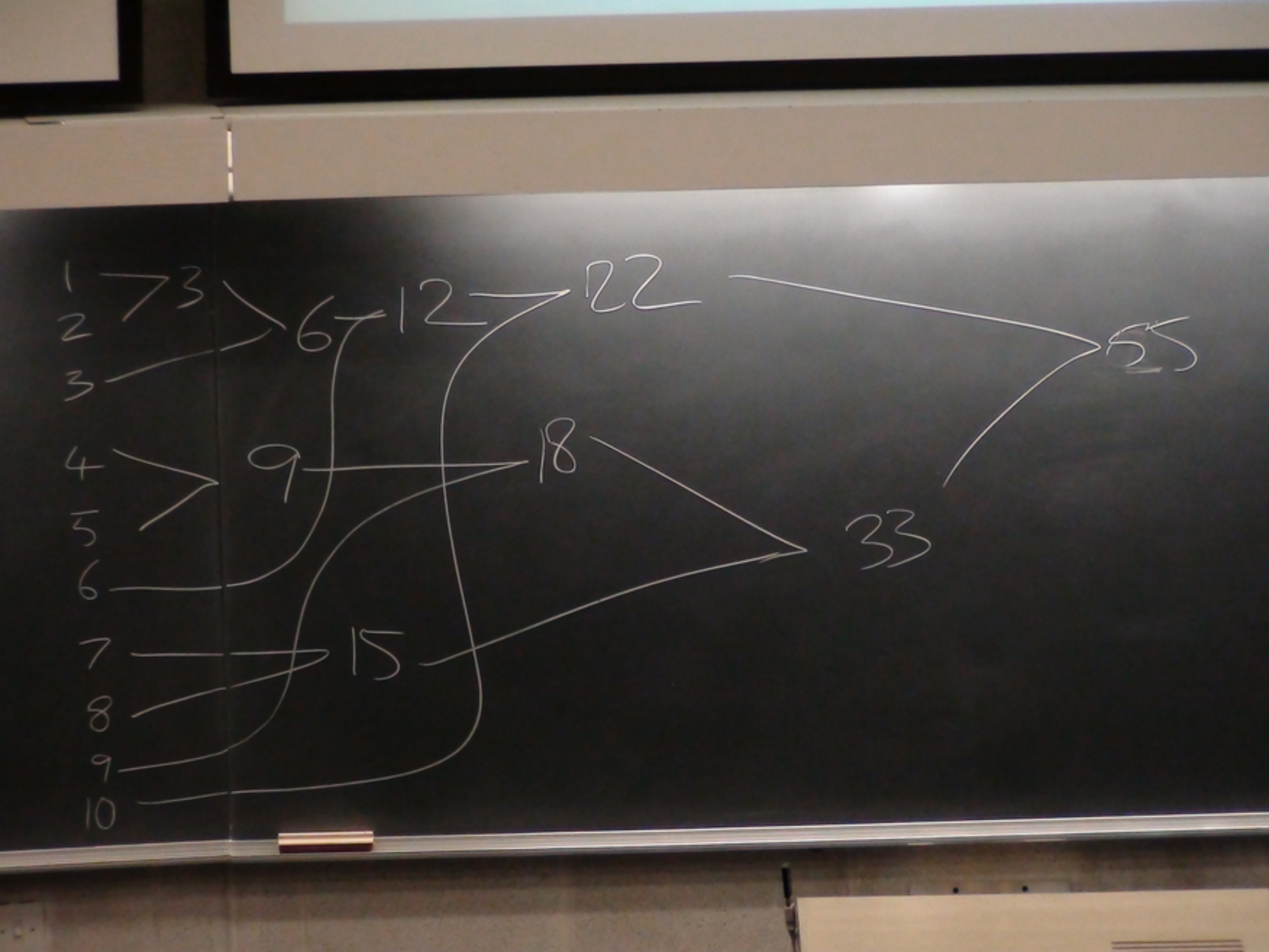
Ensemble X

a	$\frac{1}{5}$
b	$\frac{1}{5}$
c	$\frac{1}{5}$
d	$\frac{1}{5}$
e	$\frac{1}{5}$?

Ensemble Y

a	$1/Z$
b	$2/Z$
c	$3/Z$
d	$4/Z$
e	$5/Z$
f	$6/Z$
g	$7/Z$
h	$8/Z$
i	$9/Z$
j	$10/Z$?

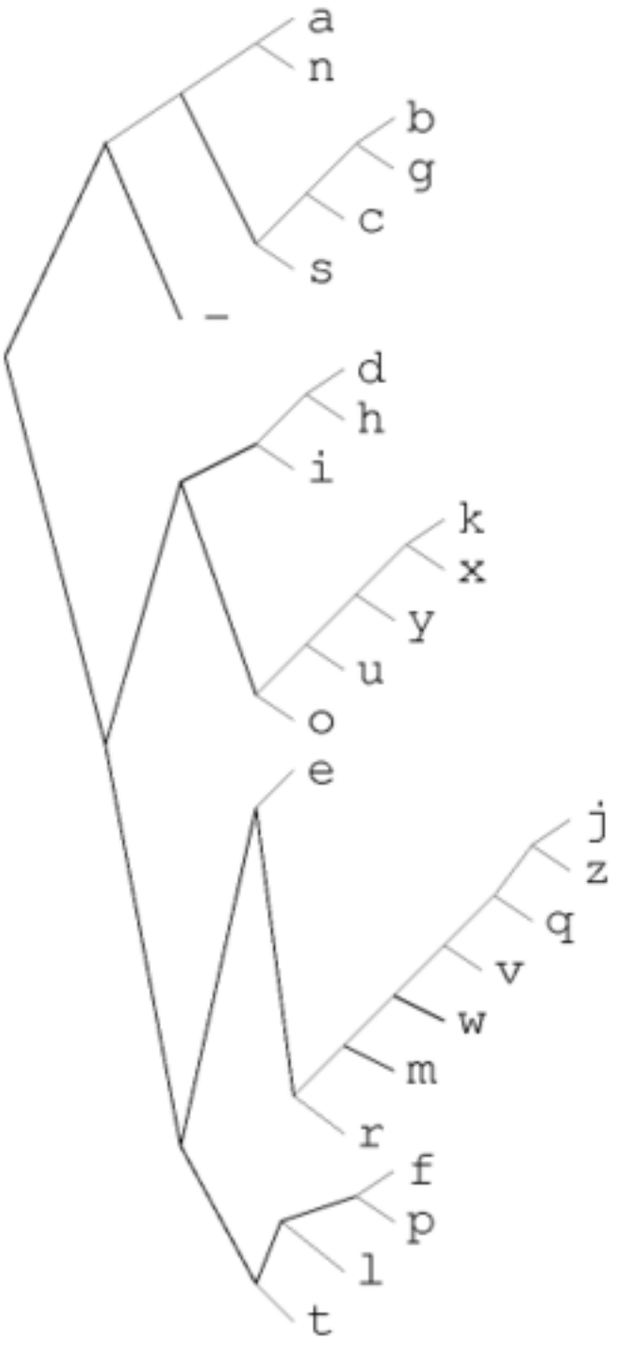

00 a 1/5
01 b 1/5
100 c 1/5
101 d 1/5
11 e 1/5
0
1
2/5
0
1
3/5
0
1
1


1
2
3
4
5
6
7
8
9
10
3
6
12
22
55
9
18
33
15

0	00	000	0000	The total symbol code budget
			0001	
		001	0010	
			0011	
	01	010	0100	
			0101	
		011	0110	
			0111	
1	10	100	1000	
			1001	
		101	1010	
			1011	
	11	110	1100	
			1101	
		111	1110	
			1111	

Prefix codes suffice

a_i	p_i	$\log_2 \frac{1}{p_i}$	l_i	$c(a_i)$
a	0.0575	4.1	4	0000
b	0.0128	6.3	6	001000
c	0.0263	5.2	5	00101
d	0.0285	5.1	5	10000
e	0.0913	3.5	4	1100
f	0.0173	5.9	6	111000
g	0.0133	6.2	6	001001
h	0.0313	5.0	5	10001
i	0.0599	4.1	4	1001
j	0.0006	10.7	10	1101000000
k	0.0084	6.9	7	1010000
l	0.0335	4.9	5	11101
m	0.0235	5.4	6	110101
n	0.0596	4.1	4	0001
o	0.0689	3.9	4	1011
p	0.0192	5.7	6	111001
q	0.0008	10.3	9	110100001
r	0.0508	4.3	5	11011
s	0.0567	4.1	4	0011
t	0.0706	3.8	4	1111
u	0.0334	4.9	5	10101
v	0.0069	7.2	8	11010001
w	0.0119	6.4	7	1101001
x	0.0073	7.1	7	1010001
y	0.0164	5.9	6	101001
z	0.0007	10.4	10	1101000001
–	0.1928	2.4	2	01

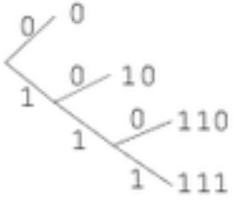


Symbol code summary

- Gibbs inequality

$$\sum_i p_i \log \frac{p_i}{q_i} \geq 0$$

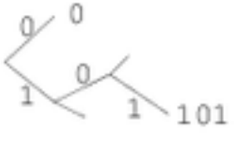
[equality only if $\mathbf{q} = \mathbf{p}$]



- Symbol codes
  - unique decodeability — requires $\sum_i 2^{-l_i} \leq 1$ Kraft inequality
  - prefix codes

    [if equality holds, the code is 'complete']
    - and binary trees

  - optimal symbol codes



The ideal codelengths l_i^* are the information contents

$$l_i^* = \log \frac{1}{p_i}$$

The optimal symbol code's expected length L satisfies

$$H(X) \leq L < H(X) + 1$$

- Huffman algorithm

Today

The ideal codelengths l_i^* are the information contents

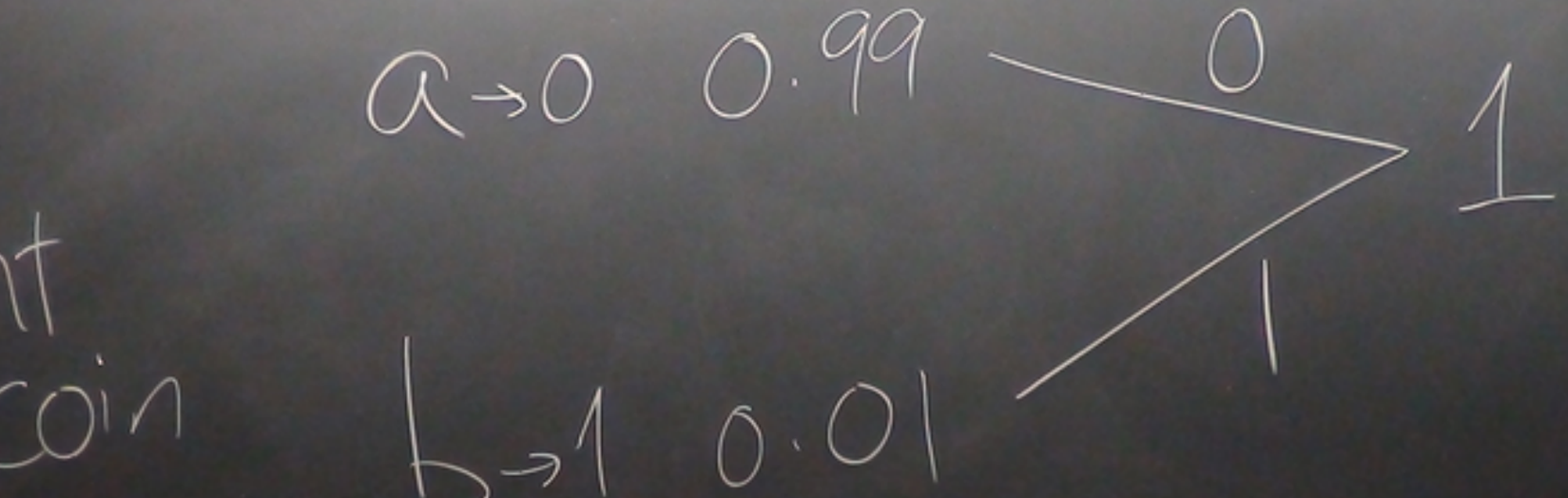
$$l_i^* = \log \frac{1}{p_i}$$

The optimal symbol code's expected length L satisfies

$$H(X) \leq L < H(X) + 1$$

- Does that wrap up compression?
  - What's wrong with optimal symbol codes?
- Arithmetic coding

Bent coin



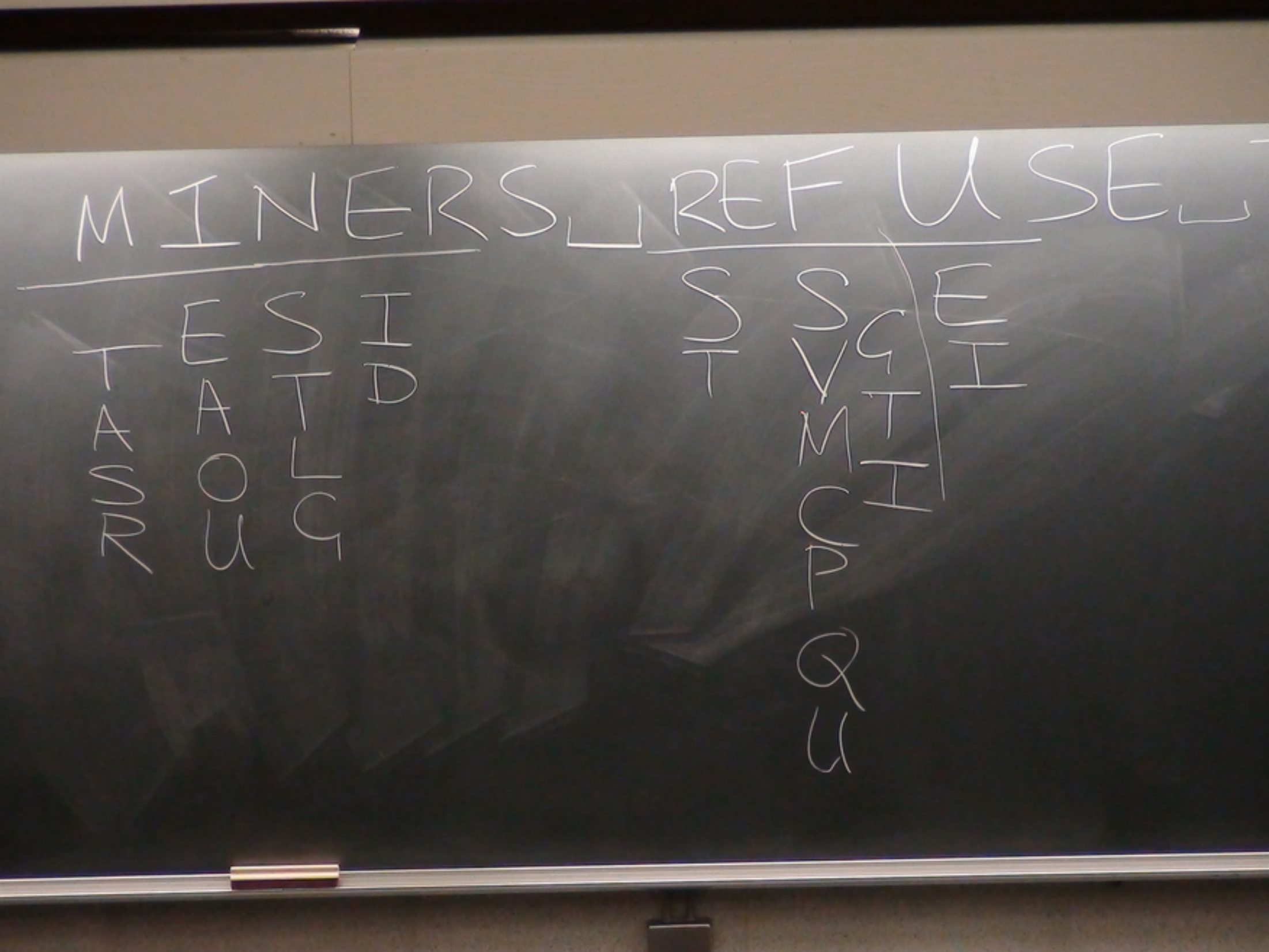
$L = 1$

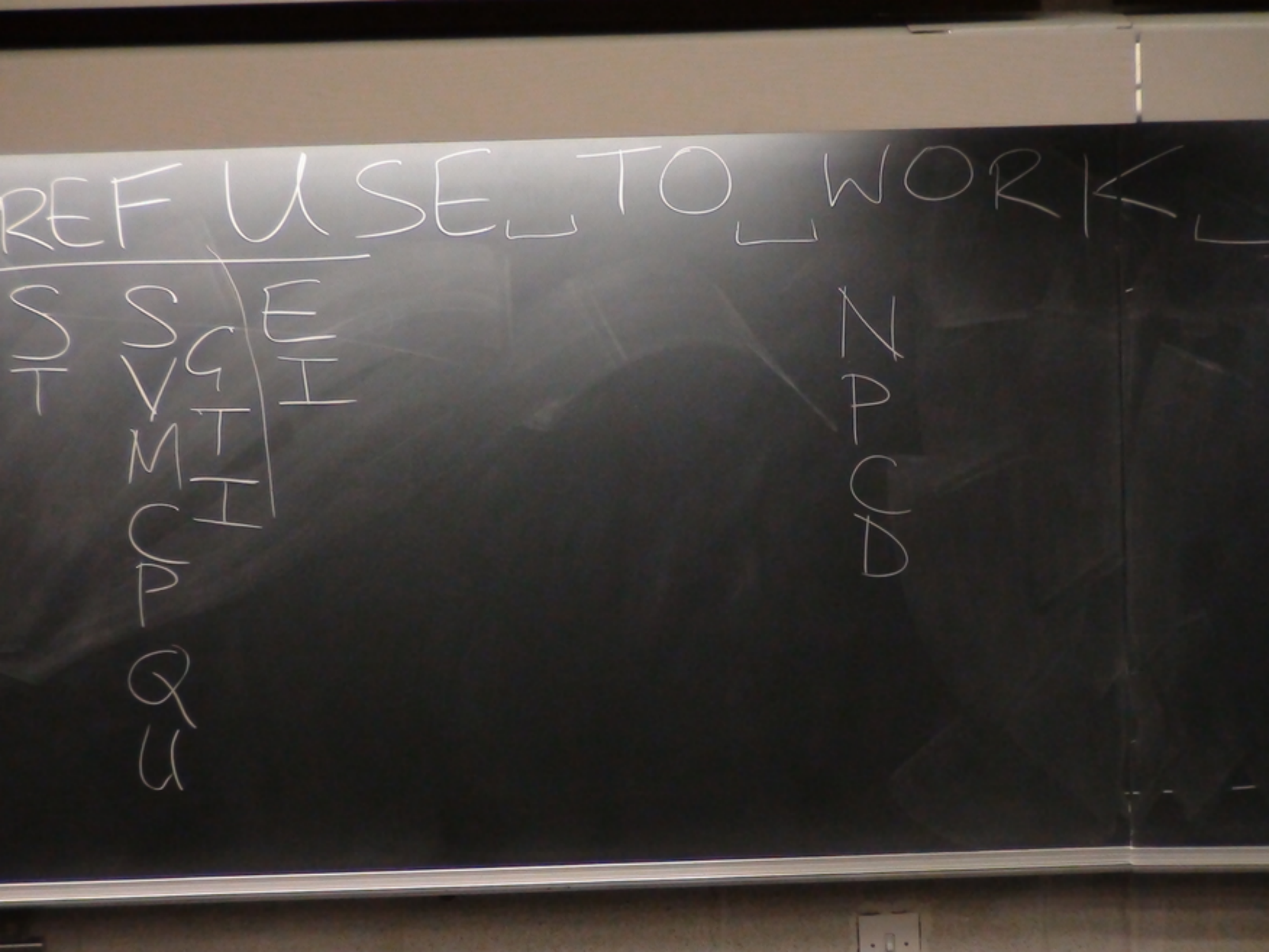
$H = 0.08$ bits

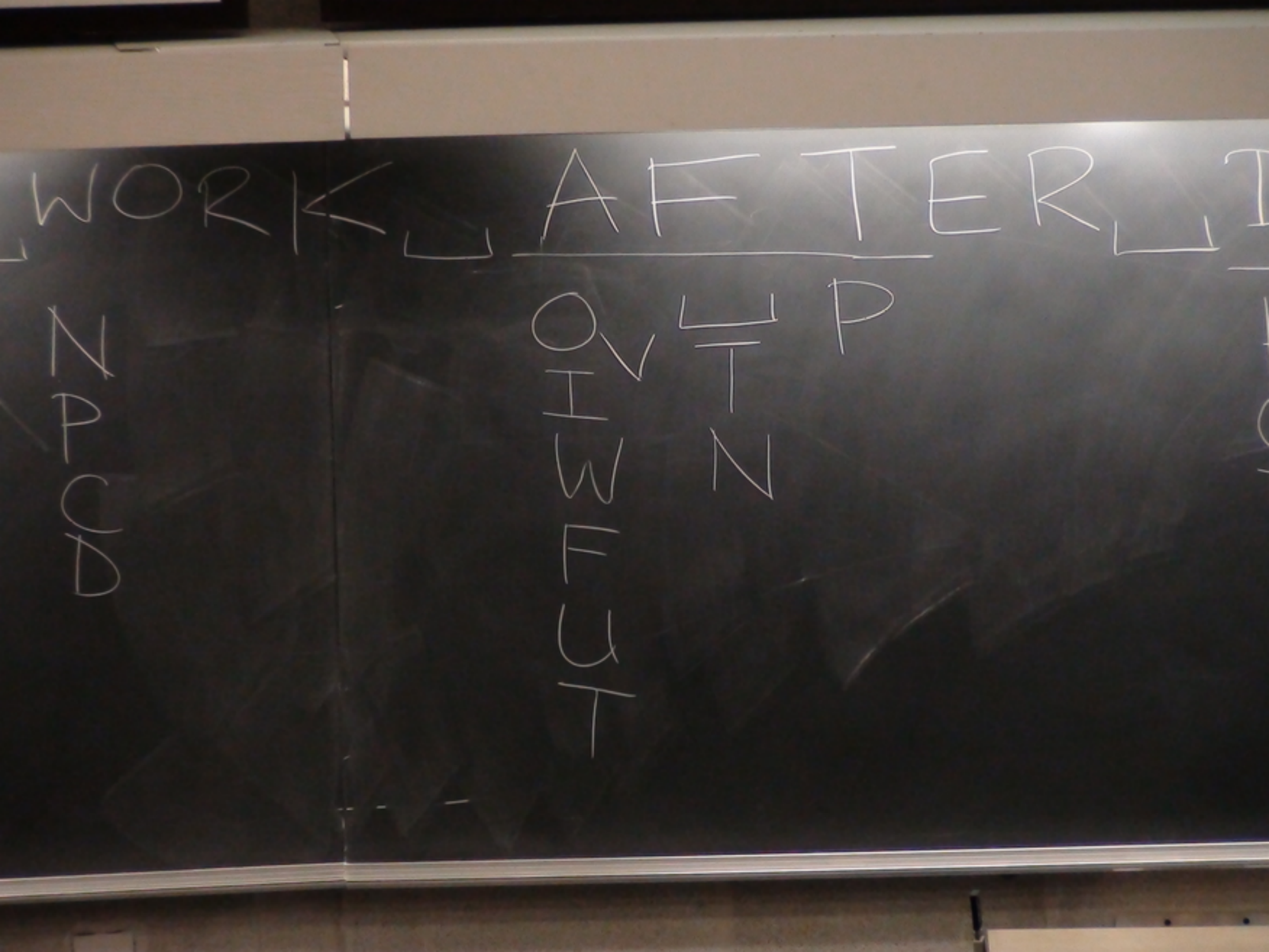
The Guessing Game

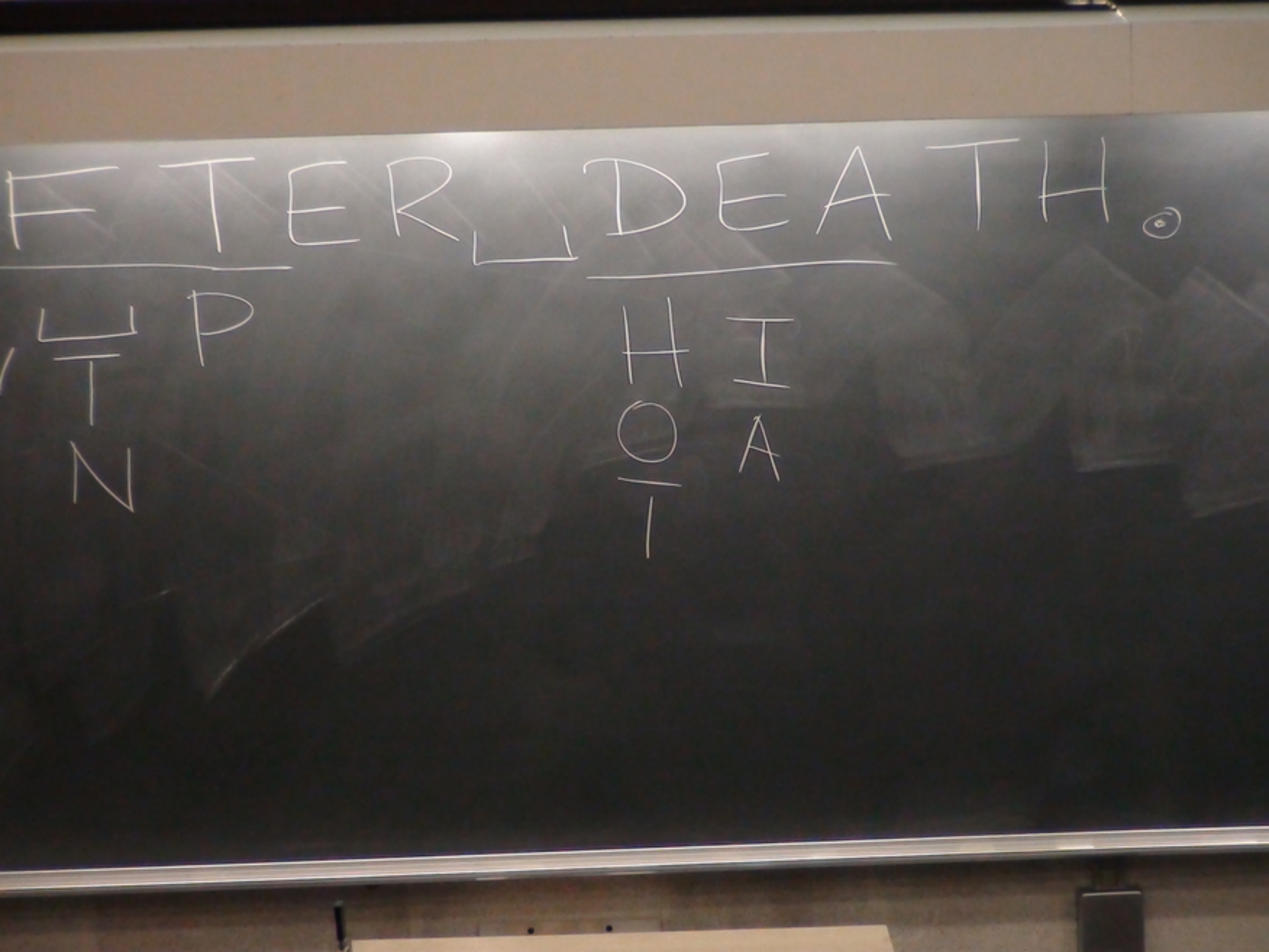
Headline composed of

{ A, B, C, D, E, F, G, H, I, J, K, L, M, . . . , Z, – }


MINERS REFUSE
TASER
TEASE
SAUCE
SIDE
STEVE
SVETLANA
MICHAEL
CHRIS
QUEEN


REFUSE TO WORK
S
T
S
V
M
C
P
Q
U
S
G
T
H
E
I
N
P
C
D

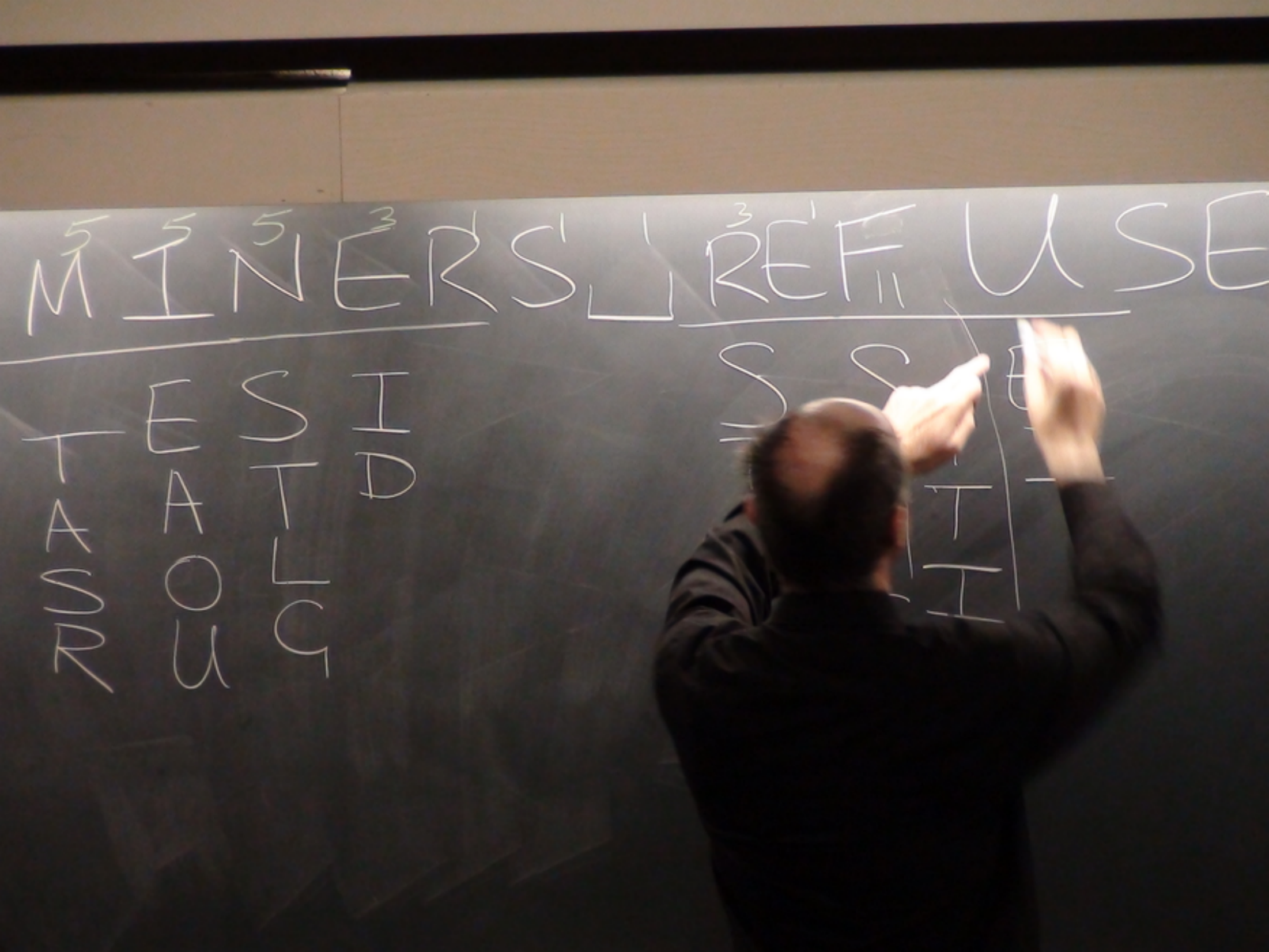

WORK
AFTER
INPUT
OUTPUT

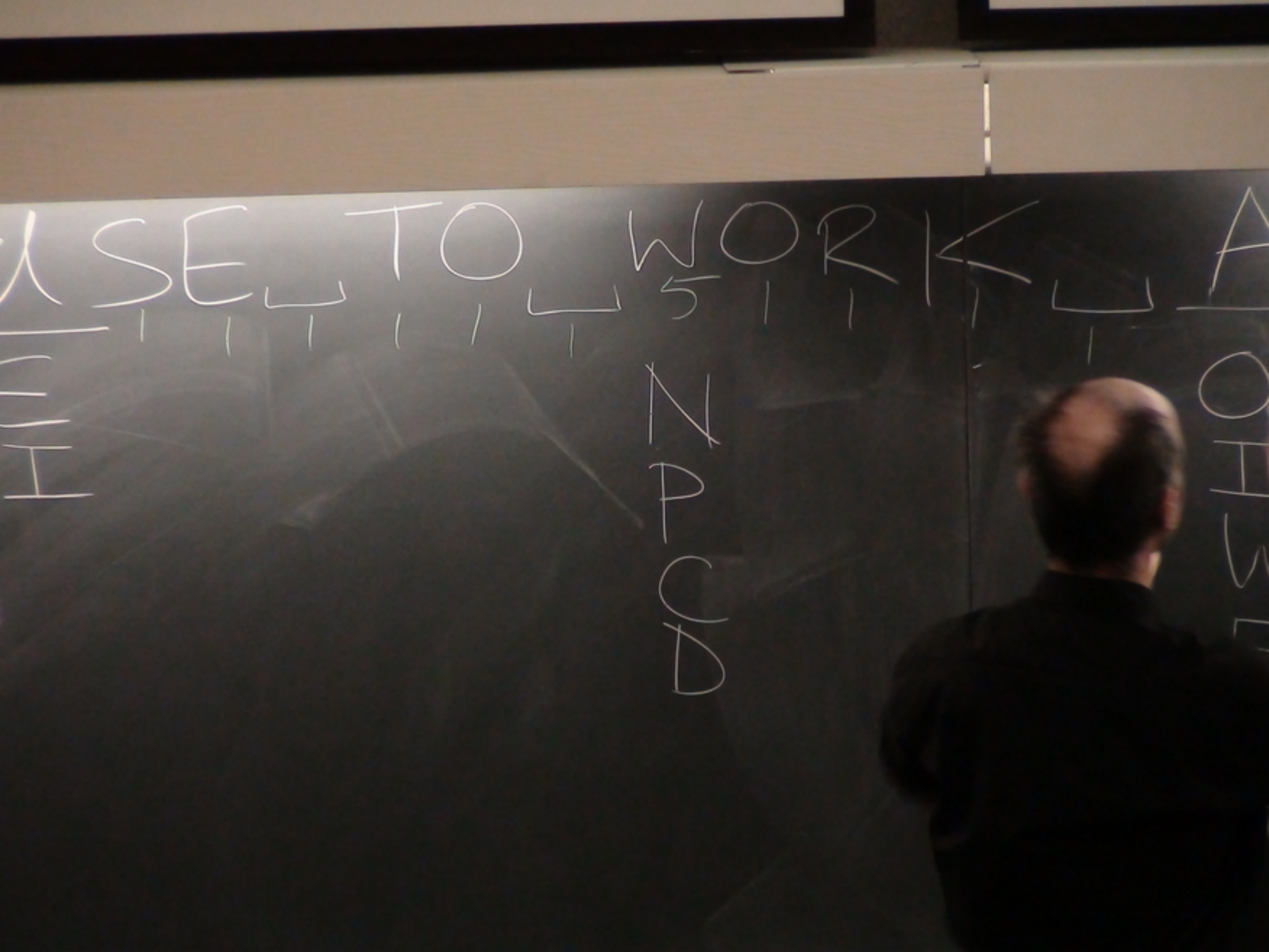

FTER DEATH.
UP
T
N
H I
O A
I

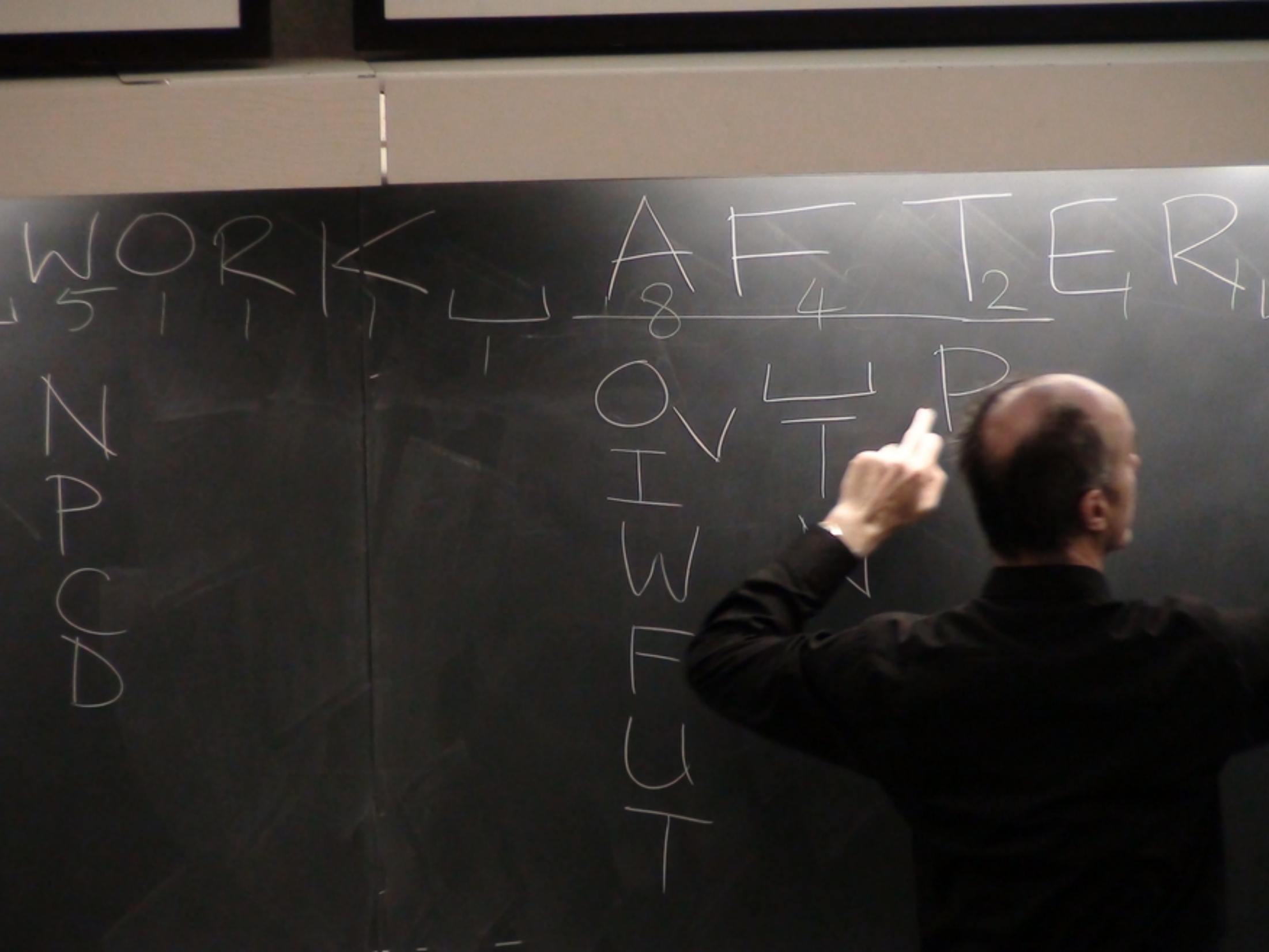
1 Predictive distribution changes with context

2 Where context-dependent predictions change as data arrive
↳ become more precise & refined

3 Often, predictive distribution is very certain


MINERS
REFUSE
TESTID
TASTAL
SOLID
RUG


USE TO WORK
NPCD


WORK
AFTER
NPCD
OVERWEIGHT

Realistic compression

Optimal symbol code for

a 0.001	00000
b 0.001	00001
c 0.990	1
d 0.001	00010
e 0.001	00011
f 0.001	0100
g 0.001	0101
h 0.001	0110
i 0.001	0111
j 0.001	0010
k 0.001	0011

expected length	1.034
entropy	0.11401
length / entropy	9

if use symbol code,

obliged to

emit $\geq \underline{1}$ bit per source character

even if $H \ll 1$

SUMMARY

symbol code,

bliged to

mit $\geq \frac{1}{}$ bit per source character

- WANT TO USE context-dependent adaptive prob'c models

even if $H \ll 1$

- Want to get rid of $(L - H)$

3 tasks

1. Bent coin

eg

a	0.99
b	0.01

Tasks

1. Bent coin

eg

a	0.99
b	0.01

2. Unknown bent coin

a	p_a
b	p_b

3. oracle

$$P(X_t \mid \underbrace{X_1 X_2 X_3 \ldots X_{t-1}}_{\text{any context}})$$

Arithmetic coding

View any binary transmission as defining an interval on the real line

$01 \equiv [0.25, 0.5]$

$100 \equiv [0.5, 0.625]$

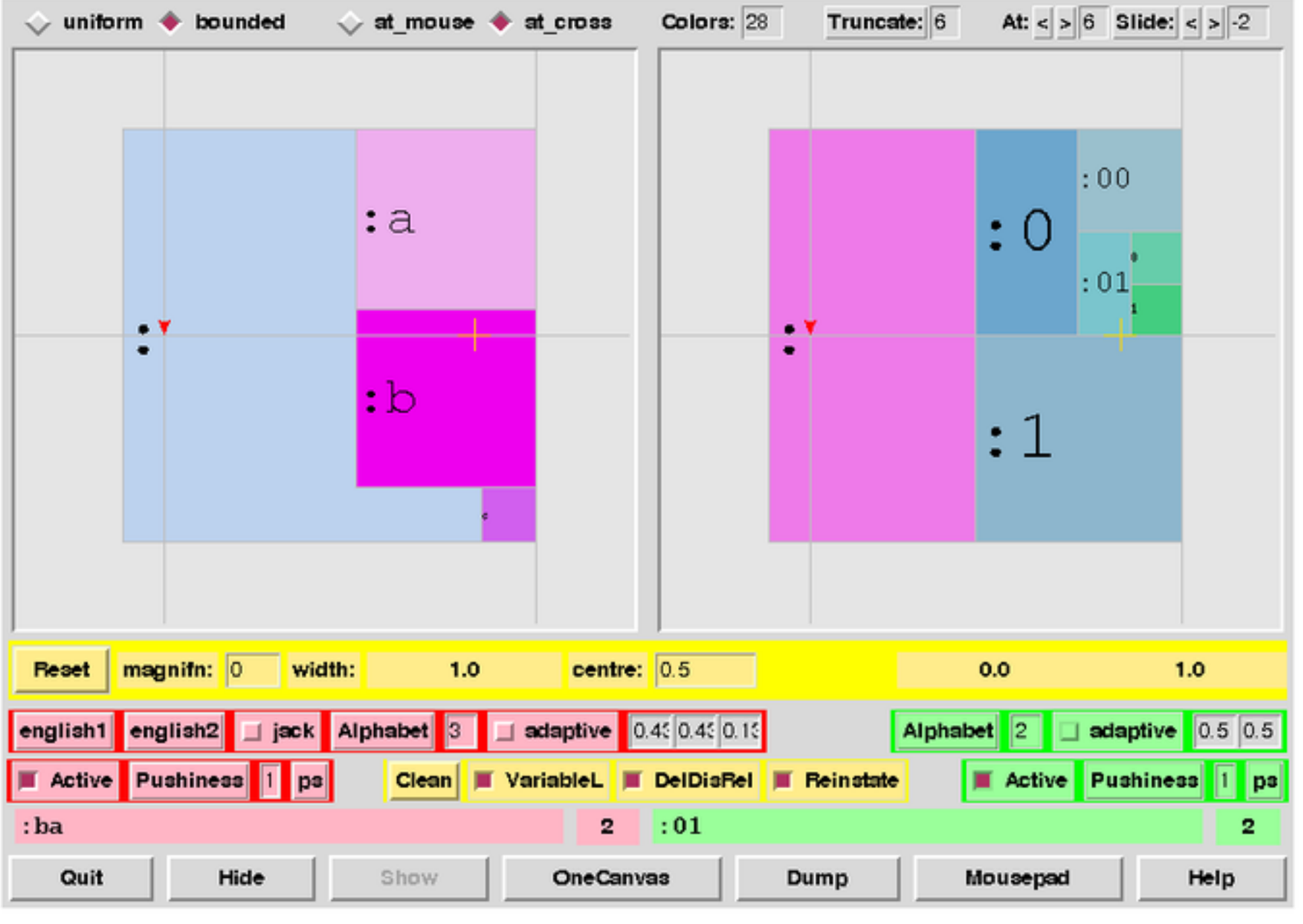
1 Mbyte file

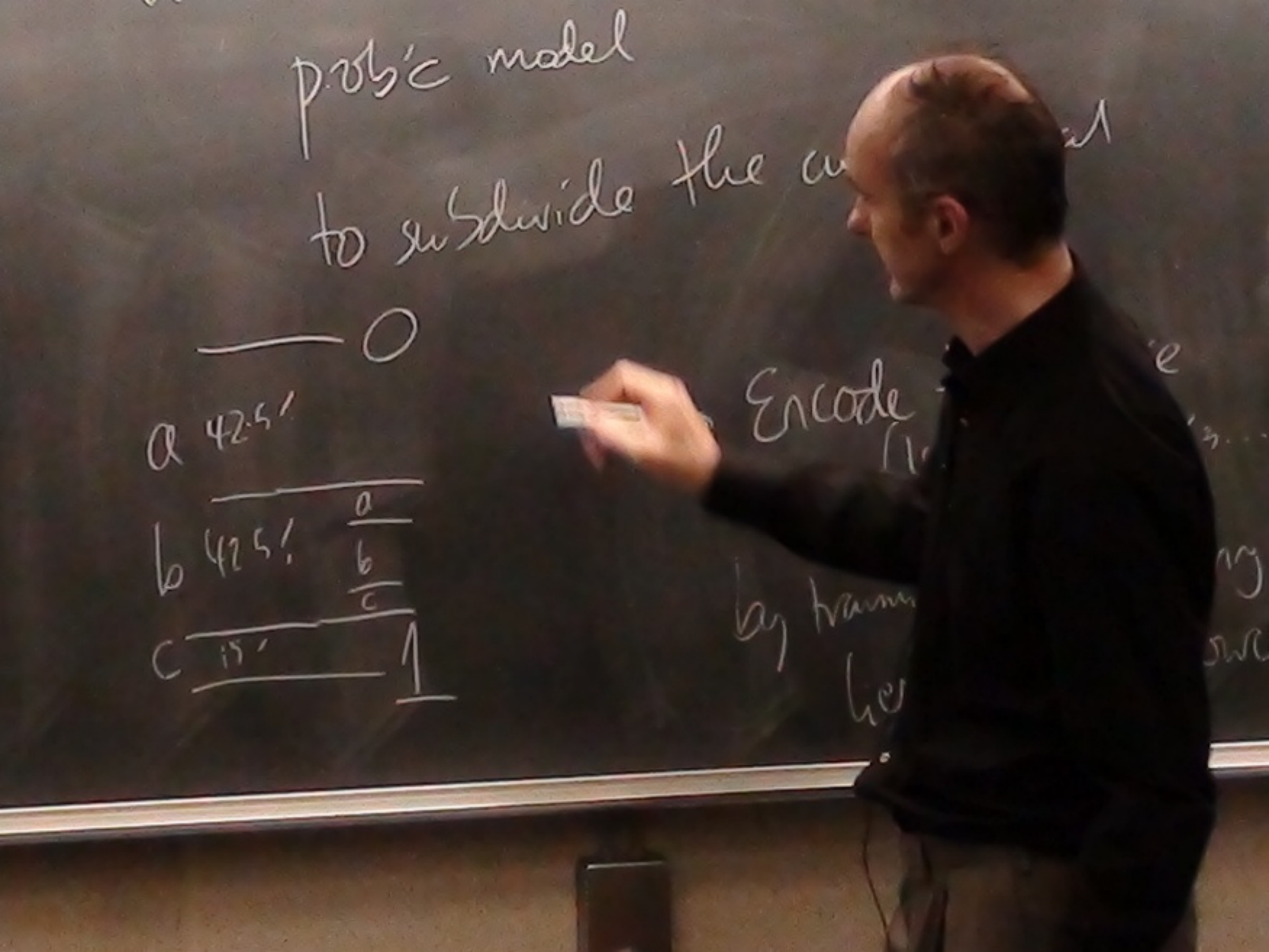
01000.....

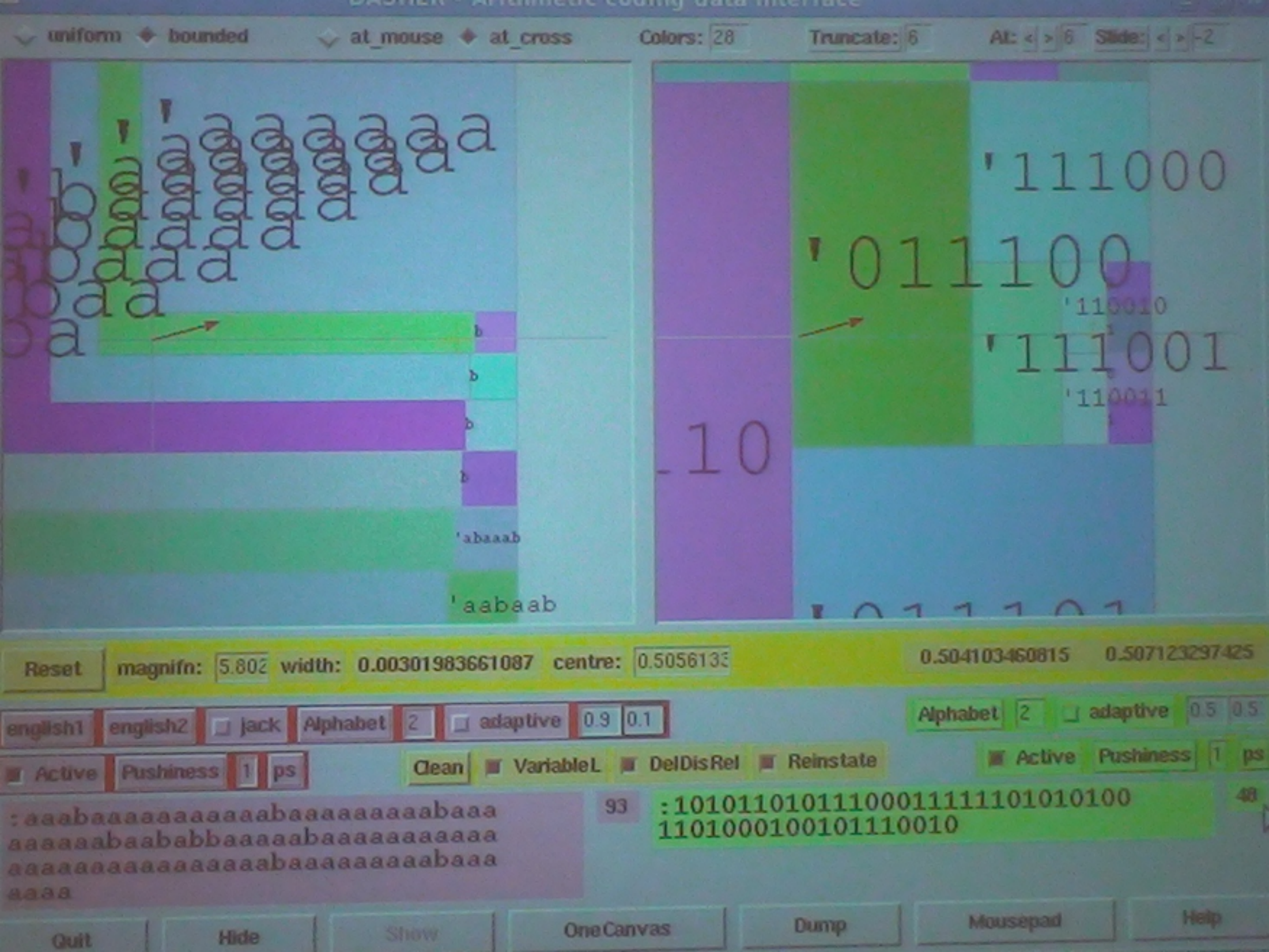
8×10^6 bits

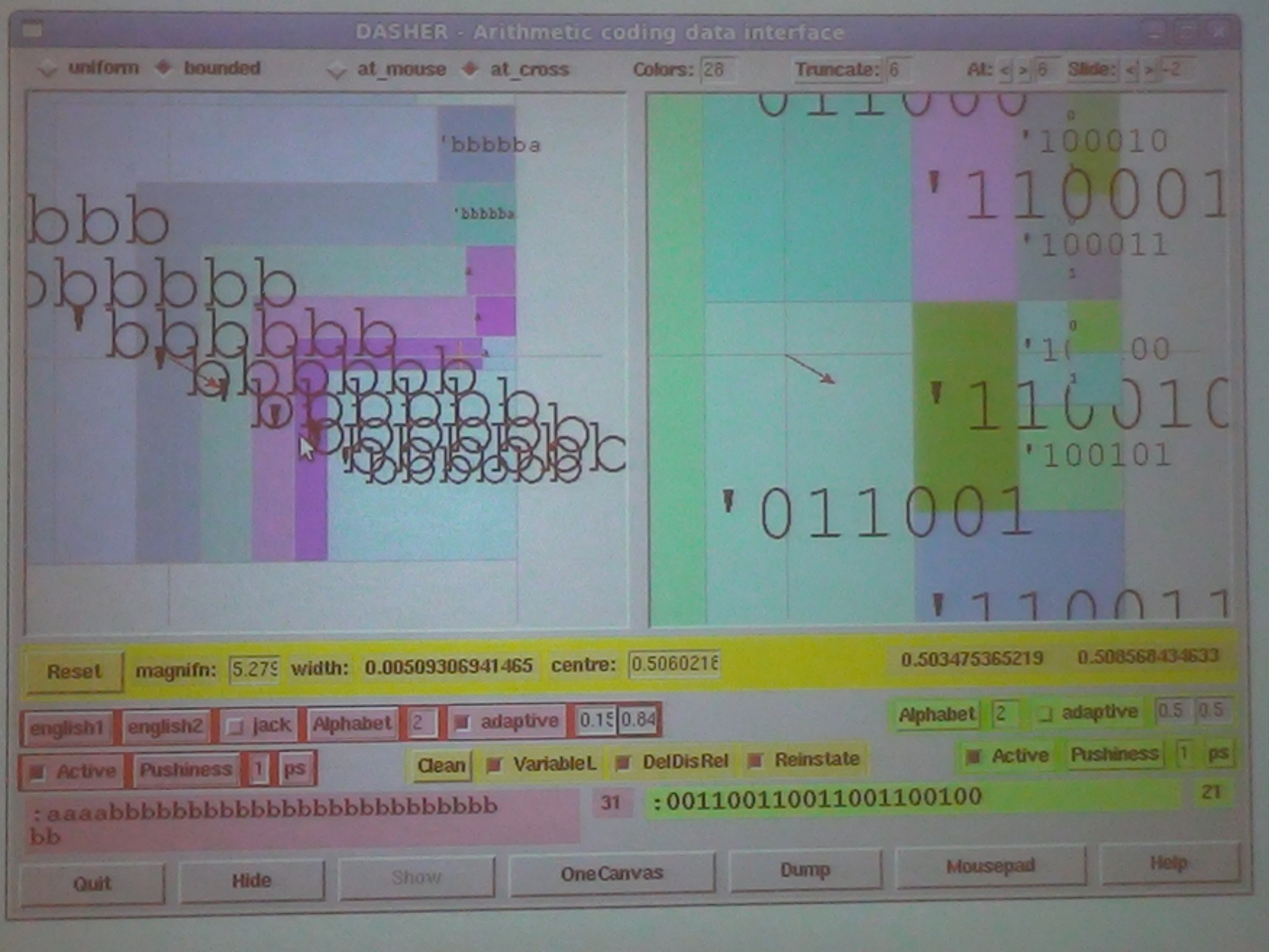
defines a real number to ~ 2 million decimal places

Arithmetic coding




prob'c model
to subdivide the
Encode
a 42.5'
b 42.5'
c 15'
a
b
c
1


uniform
bounded
at_mouse
at_cross
Colors: 28
Truncate: 6
Alt: 6
Slide: -2
'aaaaaaaa
'aaaaaaa
'aaaaaa
'aaaaa
'aaaa
'aaa
'aa
'a
'abaaab
'aabaab
'111000
'011100
'110010
'111001
'110011
-10
Reset
magnifn: 5.802
width: 0.00301983661087
centre: 0.5056133
0.504103460815
0.507123297425
english1
english2
jack
Alphabet 2
adaptive 0.9 0.1
Alphabet 2
adaptive 0.5 0.5
Active
Pushiness 1 ps
Clean
VariableL
DelDisRel
Reinstate
Active
Pushiness 1 ps
:aaabaaaaaaaaaaabaaaaaaaaaabaaa
aaaaaabaababbaaaaabaaaaaaaaaaa
aaaaaaaaaaaaaaaabaaaaaaaaabaaa
aaaa
93
:10101101011100011111101010100
11010001001011 0010
48
Quit
Hide
Show
OneCanvas
Dump
Mousepad
Help


DASHER - Arithmetic coding data interface
uniform
bounded
at_mouse
at_cross
Colors:
Truncate:
Alt:
Slide:
'bbbbba
'100010
'110001
'100011
'100101
'011001
Reset
magnifn:
width:
0.005093306941465
centre:
0.503475365219
0.508568434633
english1
english2
jack
Alphabet
adaptive
Active
Pushiness
ps
Clean
VariableL
DelDisRel
Reinstate
:aaaabbbbbbbbbbbbbbbbbbbbbbbbbb
bb
31
:00110011001100100100
21
Quit
Hide
Show
OneCanvas
Dump
Mousepad
Help

length of that binary string

$$l(x_1 \dots x_N) \leq \log_2 \frac{1}{P(x_1 \dots x_N)} + 2$$

$$P(X_1) \times P(X_2|X_1) = P(X_1, X_2)$$

$$\Rightarrow P(X_1, X_2) \times P($$

FACT 1 length of that binary

$$l(x_1, \ldots, x_N)$$

NOTE 2

length of that binary string

$$l(x_1 \dots x_N) \leq \log_2 \frac{1}{P(x_1 \dots x_N)} + 2$$

Only need to compute $N\underline{I}$ conditional probabilities

↑ alphabet

Compatible with any source alphabet & any transmitting alphabet.

used in

JBIG

PPM

DJVU

Other uses for arithmetic coding

- Efficient writing

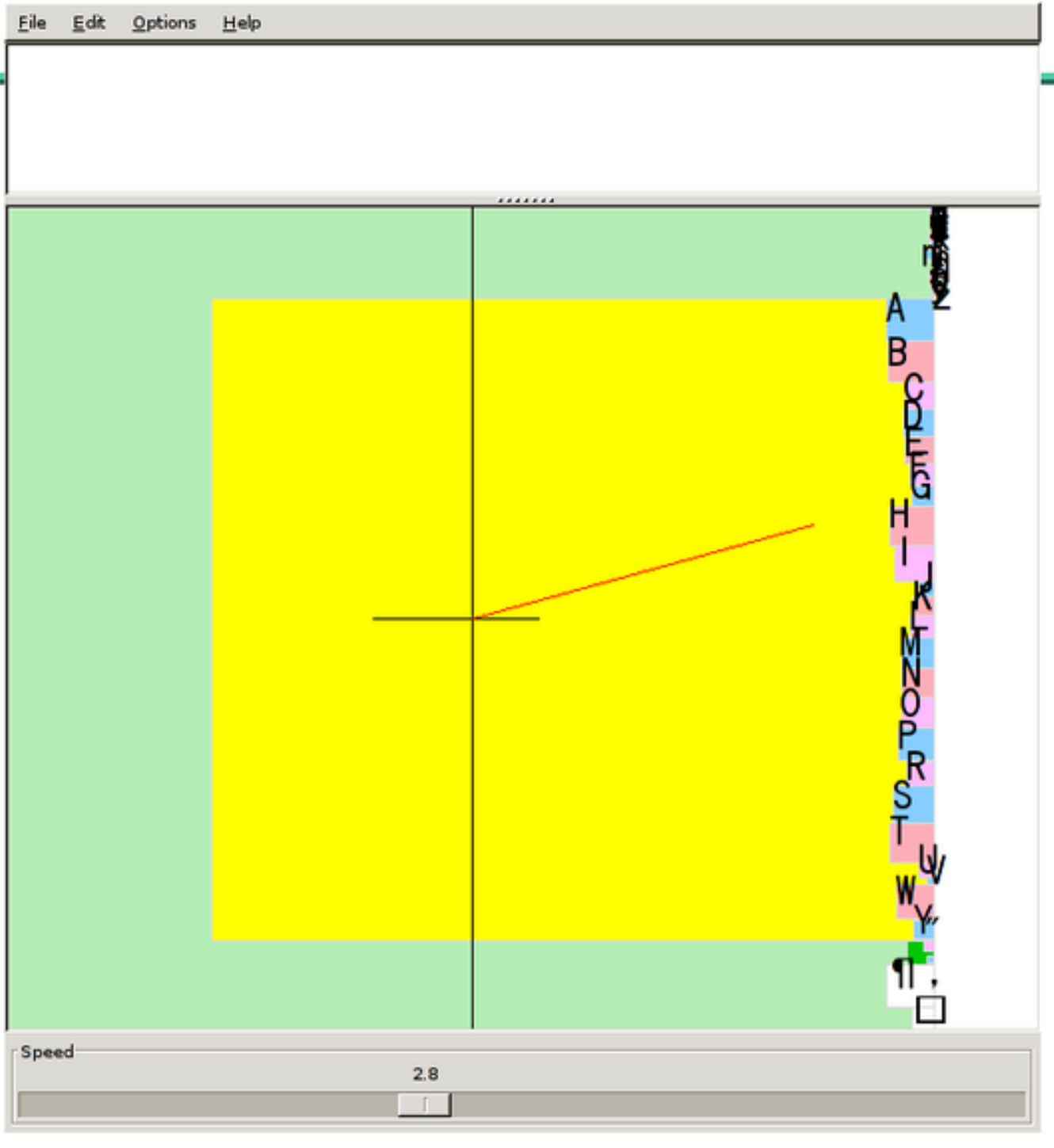
Compression:

Text $\longrightarrow$ Bit string
(preferably short)

Writing:

Text $\longleftarrow$ Gesture
(preferably brief)

Dasher



www.inference.phy.cam.ac.uk/dasher/

Q: If we encode symbols from the ensemble

$$\mathcal{A}_X = \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$$

$$\mathcal{P}_X = \{1/2, 1/4, 1/8, 1/8\}$$

using the symbol code

$$\mathcal{C} = \{\mathtt{0}, \mathtt{10}, \mathtt{110}, \mathtt{111}\},$$

what is the probability 'p_1'
that a bit plucked at random from the encoded stream
is a 1?

A $p_1 < \frac{1}{2}$

B $p_1 = \frac{1}{2}$

C $p_1 > \frac{1}{2}$

Recommended exercises

Project:

Invent a compressor and uncompressor for a source file of $N = 10,000$ bits, each having probability $f = 0.01$ of being a 1.
Implement them and/or
estimate how well your method works.

- Huffman programs huffman.p, huffman.py are on website
  - also a 'bent coin' file 0010000.... as a compression benchmark
- Also exercises 5.22, 5.26, 5.27
- Reading: Chapters 1, 2, 4, 5. And 6.
- Advance reading: Chapters 8, 9, 10